10 Two-period problem

10.1 The Budget Constraint

The first two-period problem we'll write down will be an endowment economy. Individuals will be gifted I_1 in period 1 and I_2 in period 2. In period 1, they can consume or save. In period 2, they can consume or save, but saving gives them nothing (consequently they will consume whatever they have left). If they save in period 1, then they are paid at a gross interest rate $1 + r$
The budget constraint in the first period is:

$$c_1 + s_1 = I_1$$

And in the second period is:

$$c_2 = I_2 + (1 + r)s_1$$

We can combine the two by solving for savings: $s_1 = I_1 - c_1$ and $s_1 = \frac{c_2}{1+r} - \frac{I_2}{1+r}$ and then combining them to get:

$$I_1 - c_1 = \frac{c_2}{1+r} - \frac{I_2}{1+r}$$

Rearranging:

$$\underbrace{I_1 + \frac{I_2}{1+r}}_{\text{NPV Income}} = \underbrace{c_1 + \frac{c_2}{1+r}}_{\text{NPV Expenditure}}$$

What is this saying? Why is our endowment in the second period, as well as our consumption, divided by some number? This is like when we're in an endowment economy and start out with some of good A and some of good B. We have access to some technology that converts them to a common unit (dollars, say). Mathematically, our budget constraint sells our entire endowment in the market (liquidates everything), and then buys back what we want. This budget constraint is doing the same thing. It's liquidating all our future incomes by borrowing against them. It sells all future endowments to the bank and gets the cash. Then, we buy the consumption that we want. In today's dollars, it's cheaper to buy something tomorrow than today: I only need to set aside \$1 to buy something worth \$1.01 next period at a 1% interest rate.

I depict three budget constraints in Figure 26. Each time, $I_1 = 1$ and $I_2 = 2$, but the interest rate varies: $r \in \{0, 0.1, 0.2\}$. Given I_1, I_2, and r, we have 1 equation and 2 unknowns, which we can graph out. These represent a budget constraint. When the interest rate decreases, the relative price of consumption today and consumption tomorrow increases. Say the interest rate increases. On the one hand, it's relatively more expensive to eat more today. On the other hand, it's relatively less expensive to eat more tomorrow.

10.2 Preferences

Enough about the budget constraint alone: Barro's does a great job on budget constraints. Just like our discussion of labor markets, we have preferences to be matched up with the interest rate.

We assume for now that individuals have additively separable, discounted utility across the two periods. We *could* write the Lagrangian with each budget constraint separately:

$$u(c_1, c_2) = u(c_1) + \beta u(c_2) + \lambda_1 (I_1 - c_1 - s) + \lambda_2(I_2 + (1 + r)s_1 - c_2)$$

Taking first order conditions:

$$u_{c_1} = \lambda_1$$

$$\beta u_{c_2} = \lambda_2$$

$$\lambda_1 = \lambda_2(1 + r)$$

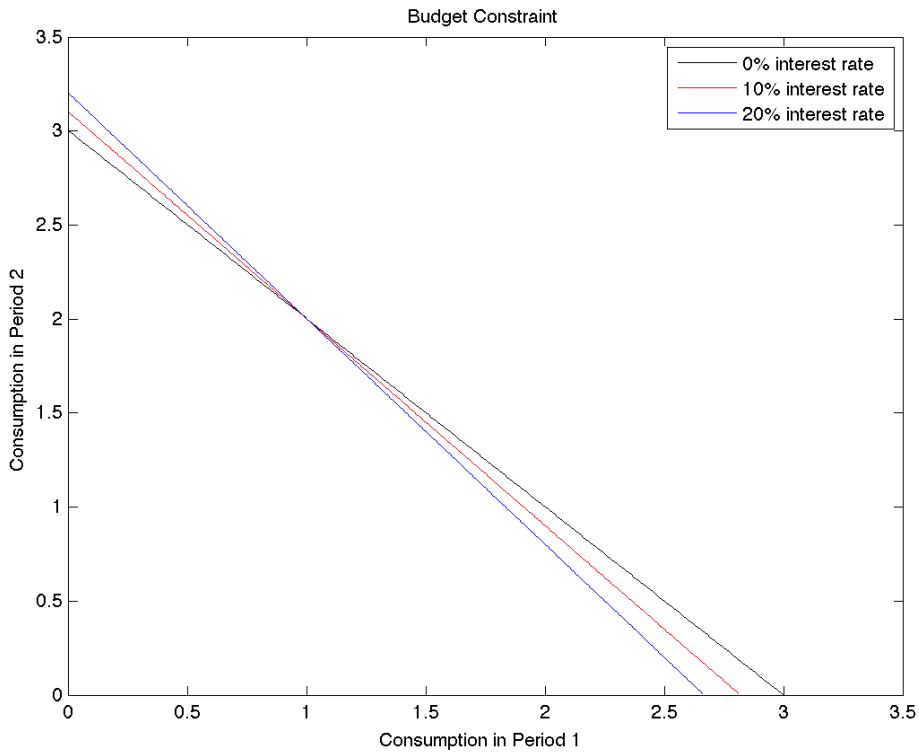

Figure 26: This is the budget set for the two-period endowment budget set above, for different interest rates. Notice that consuming the endowment in each period is always possible, no matter the interest rate: you could just boycott the lending industry, live in autarky, and consume your endowment. This is reflected in where all the lines intersect.

Remember, our Lagrange multipliers are almost always telling us something very useful. We could write:

$$\frac{1}{\beta}\frac{u_{c_1}}{u_{c_2}} = (1+r)$$

This is just like what we were doing in labor markets with labor and leisure, but now we're doing it over time. The marginal rate of substitution between consumption now and consumption later is set equal to what must be the price of transferring a dollar today to a dollar tomorrow. What does it all mean? Imagine we don't discount consumption tomorrow against consumption today, we're perfectly patient, so $\beta = 1$. Then we'd have the ratio of utilities equal to , and that certainly looks like a price, a marginal rate of technical substitution, as far as an individual is concerned. β just reflects a distortion in utility brought about by our impatience, a weighting of today as being more valuable than tomorrow.

To put some bones on the problem, we could write out:

$$L = \log(c_1) + \beta\log(c_2) + \lambda\left(I_1 + \frac{I_2}{1+r} - c_1 - \frac{c_2}{1+r}\right)$$

Solving this we get the Euler Equation:

$$\frac{c_2}{c_1} = \beta(1+r)$$

and:

$$I_1 + \frac{I_2}{1+r} = c_1 + \frac{c_2}{1+r}$$

Which give the solutions:

$$c_1^* = \frac{I_1(1+r)+I_2}{(1+r)(1+\beta)} \qquad c_2^* = \frac{\beta(I_1(1+r)+I_2)}{1+\beta}$$

Notice that the Euler Equation doesn't directly care about when you get your money. You could get it at the beginning of time, you could get it at the end. So long as the net present value of the two endowments is the same, your consumption will be the same. Put another way: the Euler Equation determines the consumption path, and the budget constraint determines the level of that path (of course, the path could depend on the level, but iI think it's helpful in this case). This is a theory of consumption, or consumption timing, and we'll talk more once we go to infinite periods. For now, let's say $\beta = 0.95$ for a time period of a year. Then we have several different paths of optimal consumption, depending on what r is by using the budget constraint and the Euler equation, we get Figure 27.

10.3 Two-Period Model with Endogenous Labor

A lot of the intuition for our Macro models can be given using very tractable two-period models that can be solved with our traditional micro tool, the Lagrangian. An individual lives for two periods, and choses how much to work in those two periods. Zero-profit firms also work for two periods, and rent labor from individuals. We'll assume in these examples that capital is given. So, we would write down:

$$u(c_t, n_t) = \log(c_1) + \psi\log(1-n_1) + \beta\log(c_2) + \beta\psi\log(1-n_2)$$

Where:

$$c_1 + \frac{c_2}{1+r_{1,2}} = w_1 n_1 + \frac{w_2}{1+r_{1,2}} n_2$$

Why does this budget constraint make sense? It's a little weird that we're dividing things that happen in the future by $(1+r_{1,2})$, right? The key is that we want to make all our decisions comparable. If use \$x today, we give up \$(1+r)x that we could have used tomorrow. Imagine that we could move dollars from today until tomorrow at a rate of $1 + r_{1,2} = \frac{1 \text{ today}}{2 \text{ tomorrow}}$. Then we can also move dollars from tomorrow until today at a rate of$1 + r_{2,1} = \frac{1}{1+r_{1,2}} = \frac{2 \text{ tomorrow}}{1 \text{ today}}$. So if we are buying consumption for tomorrow, today we would have to pay $\frac{c_2}{1+r_{1,2}}$, and if we're getting paid tomorrow, it's the equivalent of getting paid $\frac{w_2 n_2}{1+r_{1,2}}$ today.

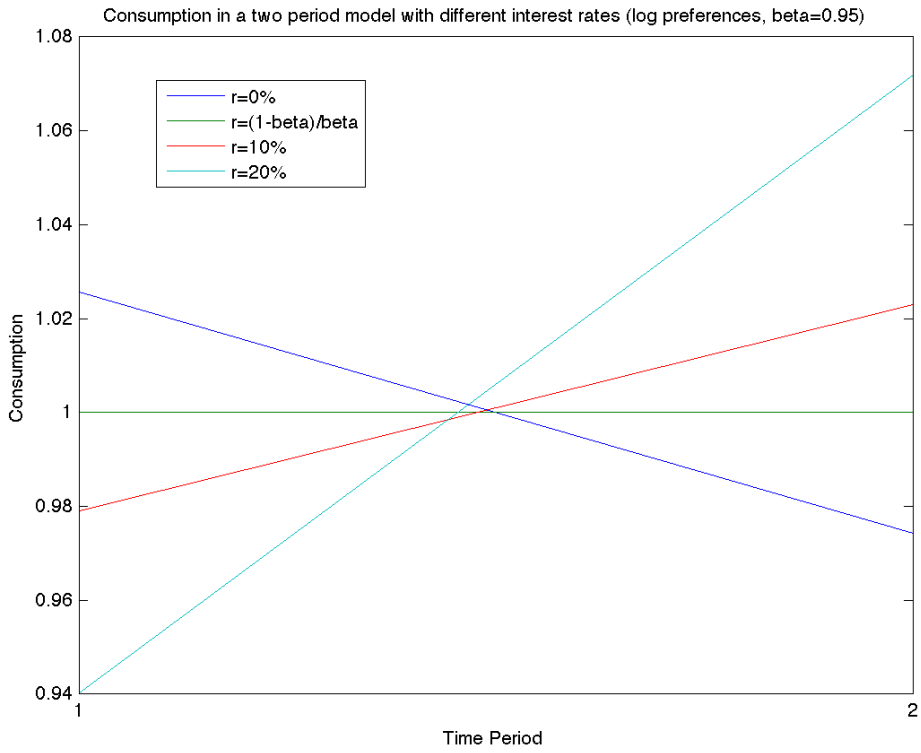

Figure 27: This graph shows four different consumption paths, depending on the interest rate, defined by the log-preference Euler equation $\frac{c_2}{c_1} = \beta(1+r)$, with $\beta = 0.95$. The Euler Equation defines the path of consumption. This will be more fun and relevant over many periods, but it tells us something, especially in regard to some of the constantly-increasing consumption we saw in chapter 1, Figure 4.

Another way of realizing this would have been to write two different budget constraints, one for today, and one for tomorrow, linking them with savings:

$$c_1 = w_1 n_1 - s_1 \Rightarrow s_1 = w_1 n_1 - c_1$$
$$c_2 = w_2 n_2 + (1 + r_{1,2}) s_1 \Rightarrow s_1 = \frac{c_2}{(1 + r_{1,2})} - \frac{w_2 n_2}{(1 + r_{1,2})}$$

Combining the two by getting rid of s_1 we get back our original budget constraint:

$$w_1 n_1 - c_1 = \frac{c_2}{(1 + r_{1,2})} - \frac{w_2 n_2}{(1 + r_{1,2})}$$

Which we could write:

$$c_1 + \frac{c_2}{(1 + r_{1,2})} = w_1 n_1 + \frac{w_2 n_2}{(1 + r_{1,2})}$$

So, we have:

$$L = \log(c_1) + \psi \log(1 - n_1) + \beta \log(c_2) + \beta\psi \log(1 - n_2) + \lambda \left(w_1 n_1 + \frac{w_2 n_2}{(1 + r_{1,2})} - c_1 - \frac{c_2}{(1 + r_{1,2})} \right)$$

Then we choose c_1, n_1, c_2, n_2, and λ. Taking FOC's:

$$\begin{aligned}
c_1 :& \frac{1}{c_1} - \lambda = 0 \\
c_2 :& \beta \frac{1}{c_2} - \lambda \frac{1}{1 + r_{1,2}} = 0 \\
n_1 :& -\psi \frac{1}{1 - n_1} + \lambda w_1 = 0 \\
n_2 :& -\beta\psi \frac{1}{1 - n_2} + \lambda \frac{w_2}{1 + r_{1,2}} = 0 \\
\lambda :& c_1 + \frac{c_2}{(1 + r_{1,2})} = w_1 n_1 + \frac{w_2 n_2}{(1 + r_{1,2})}
\end{aligned}$$

Solving this system algebraically:

$$\begin{aligned}
c_1^* &= \frac{w_1(1 + r) + w_2}{(1 + r)(1 + \beta)(1 + \psi)} \\
c_2^* &= \frac{\beta(w_1(1 + r) + w_2)}{(1 + \beta)(1 + \psi)} \\
n_1^* &= \frac{w_1(1 + r)(1 + \beta(1 + \psi)) - w_2\psi}{(1 + r)(1 + \beta)(1 + \psi)w_1} \\
n_2^* &= \frac{w_2(1 + \beta + \psi) - w_1\beta\psi(1 + r)}{(1 + \psi)(1 + \beta)w_2} \\
\lambda^* &= \frac{(1 + r)(1 + \beta)(1 + \psi)}{(1 + r)w_1 + w_2}
\end{aligned}$$

What do we get from this? We have several "parameters" or "givens" we're interested in: β, w_1, w_2, r, and ψ. The individual takes all these as given, and we're interested in the effects of changing each one. I'll make a table of all the first-order derivatives.

Log preferences specifically give us something of interest. Let's remind ourselves of what: if I increase or decrease both w_1 and w_2 by the same amount, say by α, so that $w_1 \to \alpha w_1$ and $w_2 \to \alpha w_2$ then I would have:

$$
\begin{aligned}
c_1^{new} &= \frac{\alpha w_1(1+r)+\alpha w_2}{(1+r)(1+\beta)(1+\psi)} = \alpha c_1^* \\
c_2^{new} &= \frac{\beta(\alpha w_1(1+r)+\alpha w_2)}{(1+\beta)(1+\psi)} = \alpha c_2^* \\
n_1^{new} &= \frac{\alpha w_1(1+r)(1+\beta(1+\psi))-\alpha w_2\psi}{(1+r)(1+\beta)(1+\psi)\alpha w_1} = n_1^* \\
n_2^{new} &= \frac{\alpha w_2(1+\beta+\psi)-\alpha w_1\beta\psi(1+r)}{(1+\psi)(1+\beta)\alpha w_2} = n_2^* \\
\lambda^{new} &= \frac{(1+r)(1+\beta)(1+\psi)}{(1+r)\alpha w_1+\alpha w_2} = \frac{\lambda^*}{\alpha}
\end{aligned}
$$

This reminds us of something we already knew: that an increase in (all) wages will have an income effect away from working, and a substitution effect toward working, which exactly cancel in log utility.

Comparative Statics: $\frac{\partial X}{\partial Y}$, $X \in \{c_1, c_2, n_1, n_2, \lambda\}$, $Y \in \{\beta, w_1, w_2, r, \psi\}$

	c_1	c_2	n_1	n_2	λ
$\frac{\partial X}{\partial \beta}$	$-\frac{(1+r)w_1+w_2}{(1+\beta)^2(1+r)(1+\psi)}$	$\frac{(1+r)w_1+w_2}{(1+\beta)^2(1+\psi)}$	$\frac{\psi((1+r)w_1+w_2)}{(1+\beta)^2(1+r)w_1(1+\psi)}$	$-\frac{\psi((1+r)w_1+w_2)}{(1+\beta)^2w_2(1+\psi)}$	$\frac{(1+r)(1+\psi)}{(1+r)w_1+w_2}$
$\frac{\partial X}{\partial w_1}$	$\frac{1}{1+\psi+\beta+\psi\beta}$	$\frac{\beta+\beta r}{1+\psi+\beta+\psi\beta}$	$\frac{w_2\psi}{(1+\beta)(1+r)w_1^2(1+\psi)}$	$-\frac{\beta(1+r)\psi}{(1+\beta)\mathrm{w}2(\psi+1)}$	$-\frac{(1+\beta)(1+r)^2(1+\psi)}{((1+r)w_1+w_2)^2}$
$\frac{\partial X}{\partial w_2}$	$\frac{1}{(1+\beta)(1+r)(1+\psi)}$	$\frac{\beta}{1+\psi+\beta+\psi\beta}$	$-\frac{\psi}{(1+\beta)(1+r)w_1(1+\psi)}$	$\frac{\beta(1+r)w_1\psi}{(1+\beta)w_2^2(1+\psi)}$	$-\frac{(1+\beta)(1+r)(1+\psi)}{((1+r)w_1+w_2)^2}$
$\frac{\partial X}{\partial r}$	$-\frac{w_2}{(1+\beta)(1+r)^2(1+\psi)}$	$\frac{\beta w_1}{1+\psi+\beta+\psi\beta}$	$\frac{w_2\psi}{(1+\beta)(1+r)^2w_1(1+\psi)}$	$-\frac{\beta w_1\psi}{\beta w_2\psi+\beta w_2+w_2\psi+w_2}$	$\frac{(1+\beta)w_2(1+\psi)}{((1+r)w_1+w_2)^2}$
$\frac{\partial X}{\partial \psi}$	$-\frac{(1+r)w_1+w_2}{(1+\beta)(1+r)(1+\psi)^2}$	$-\frac{\beta((1+r)w_1+w_2)}{(1+\beta)(1+\psi)^2}$	$-\frac{(1+r)w_1+w_2}{(1+\beta)(1+r)w_1(1+\psi)^2}$	$-\frac{\beta((1+r)w_1+w_2)}{(1+\beta)w_2(1+\psi)^2}$	$\frac{(1+\beta)(1+r)}{(1+r)w_1+w_2}$

Static	Interpretation of Increase	c_1	c_2	n_1	n_2	λ
$\frac{\partial X}{\partial \beta}$	More Patient	(-)	(+)	(+)	(-)	(+)
$\frac{\partial X}{\partial w_1}$	First period wage increase	(+)	(+)	(+)	(-)	(-)
$\frac{\partial X}{\partial w_2}$	Second period wage increase	(+)	(+)	(-)	(+)	(-)
$\frac{\partial X}{\partial r}$	Increase in interest rate	(-)	(+)	(+)	(-)	(+)
$\frac{\partial X}{\partial \psi}$	Like work less	(-)	(-)	(-)	(-)	(+)